

G02 / PROTECTED ANSWERS

G02 REVIEW B

Standalone Zero-to-Job-Ready Data Engineer 2026

Study, practise, check and explain.

A complete learning route

Start with the lesson video. Read the worked explanation beside it, pause to reproduce the example, then solve the supplied task. The books contain your study notes; keep your own work and results as evidence.

Use the protected answers after saving your first attempt. A changed retry shows what you can now do independently. There is no forced timer during learning.

One course. M00-M19. 264 lessons and 50 project phases.

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Read this first

Open this book after saving your own attempt. Compare the reasoning and result, repair the cause of a mistake, then close the answer and try changed inputs.

YOUR STANDALONE GATE BOOK

Open only after Gate B attempt is saved

Assessment rule: Save the current attempt before opening its review. Fresh Gate B stays closed until explicitly assigned after repair.

Independent source controls

The source contains 4 customers, 5 orders and 6 item lines. Item gross is 300. Original order-header fees total 33. Therefore all-order gross must reconcile to 333.

Correct order-grain result

The five orders are:

Order Customer Region group

Payment Item gross Fee Order gross

Q701	R01	East	Paid	100	10	110
Q702	R02	West	Unpaid	60	10	70
Q703	R01	East	Unknown	50	5	55
Q704	R03	North	Paid	90	8	98
Q705	(missing)	Unknown	Unpaid	0	0	0

The order result has five rows and five distinct order IDs. Items 300 + fees 33 = gross 333. Q705 must survive even though it has no items and no customer identity.

Raw-join failure

An inner join from orders directly to item lines has six rows and represents only four orders. Header fees repeat with item lines: Q701's 10 appears twice and Q704's 8 appears twice, giving a joined fee sum of 51, not 33. SUM(DISTINCT shipping_fee) is also wrong: distinct values 10, 5, 8 and 0 sum to 23, because equal numeric fees on different real orders are not duplicate records.

Region reasoning

East: two orders, gross 165, mean 82.50. West: one order, 70, mean 70. North: one order, 98, mean 98. Unknown: one order, 0, mean 0. Regions with gross ≥ 90 are East and North, total 263.

The overall mean is $333 / 5 = 66.60$. The simple unweighted average of the four region means is $(82.5 + 70 + 98 + 0) / 4 = 62.625$. The difference exists because regions contain unequal numbers of orders.

Highest versus latest

For R01, highest is Q701/110 while latest is Q703 on 2026-12-11. R02 has Q702 for both; R03 has Q704 for both. This proves that value ordering and date ordering answer different questions. Keep a deterministic secondary order even if this fixture does not force a tie.

Set and absence logic

Selecting customer IDs from Paid orders and appending Unknown-payment orders with UNION ALL yields R01 twice and R03 once. UNION yields R01 and R03 once each. Customer R04 has no orders and should be found with a null-safe NOT EXISTS pattern.

Date boundary

The 2026-12-10 population is Q701 and Q702. A robust day predicate is `ordered_on >= DATE '2026-12-10' AND ordered_on < DATE '2026-12-11'`. The half-open form extends safely to timestamps because the next-day boundary is excluded.

PostgreSQL evidence

A correct disposable contract should reject at least one intentionally invalid row and preserve the error you actually observed. An explicit transaction should show a changed value before ROLLBACK and the original afterward. EXPLAIN is an estimated plan; EXPLAIN ANALYZE executes and reports observed work. On these tiny tables, do not claim an index is "faster" merely because one plan node appears.

Repair route

Use this review only after the saved Gate B attempt. Repair the specific weak dimension, close the review, and prove the same competency on changed data or a different disposable object. If multiple dimensions remain review-dependent, return to M02/M03 practice instead of repeatedly rereading this answer.